Fictive Kinship Vocatives as Memetic Technology

An Evolutionary Analysis of "Primo", "Compadre", and "Brother"

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Abstract

This paper proposes a functional hypothesis about address terms derived from kinship vocabulary: expressions such as "primo" (cousin), "hermano" (brother), "compadre," and "bro," directed at genetically unrelated interlocutors, are not superficial vernacular conventions but *low-entropy memes* that activate evolutionarily conserved cognitive modules, reduce friction costs in low-trust environments, and function as coalitional signaling devices. The argument integrates three theoretical frameworks: Dawkins's extended phenotype theory and memetics, Henrich's comparative cultural psychology (WEIRD/non-WEIRD distinction), and multilevel evolutionary game theory (EGT). Ethnographic evidence from the ju/'hoansi, the Ache of Paraguay, and the Matsiguenka of Amazonia documents the mechanism directly. The paper analyzes a formalization gradient running from momentary colloquial activation (near-zero cost, no codified obligations) to the full Catholic compadrazgo institution (ritual entry cost, lifelong reciprocal obligations), with intermediate cases in between. Three functions are identified in permanent tension: horizontal leveling, clientelar lubrication, and semantic hollowing through appropriation. The central prediction is that the frequency of fictive kinship vocatives directed at strangers covaries negatively with institutional quality: where abstract institutions fail to produce impersonal trust, agents recruit relational substitutes, and the fictive kinship vocative is the cheapest available option.

Keywords: extended phenotype, memetics, evolutionary game theory, Henrich, WEIRD, vocatives, fictive kinship, compadrazgo, institutional trust, AAVE, social capital, kin selection

1. The Field Observation

A Venezuelan addresses a taxi driver he has just met as "primo." A Mexican asks a favor from a neighbor calling him "compa." A Black American from Chicago greets a churchmate with "brother." All three deploy a term that, in its literal sense, denotes a genetic or quasi-genetic bond. None of them shares that bond with their interlocutor.

The obvious interpretation is that these are semantically empty conventions: residual phonological forms stripped of their original referential content. This paper argues the contrary. The terms retain, precisely because of their origin in kinship registers, a functional cognitive load that operates below the speaker's threshold of conscious awareness. The vocative is a low-cost technology that exploits the receiver's evolved psychology.

The triggering observation is geographically specific: the use of "primo" and "hermano" in Venezuela and the Spanish-speaking Caribbean as vocatives directed at strangers is systematically more frequent than in the Southern Cone. That asymmetry is not trivial. If these were mere regional mannerisms without adaptive function, their geographic distribution should be random or explicable by superficial cultural diffusion. The alternative hypothesis, defended here, is that their frequency covaries with institutional weakness: where the state fails to produce impersonal trust, agents fall back on relational substitutes, and the fictive kinship vocative is the most economical available substitute.

The phenomenon is not limited to informal colloquial address. At one end of the formalization spectrum, "primo" to a stranger is momentary and costs nothing. At the other end, the Catholic compadrazgo institution is a formally codified fictive kinship system with ritual entry costs and lifelong reciprocal obligations. The colloquial vocative and the full institution share the same underlying mechanism: activation of kin-detection cognitive modules in the absence of genetic kinship. Understanding that shared mechanism is the central task of this paper.

2. Theoretical Framework

2.1. Extended Phenotype and Memetics

In *The Extended Phenotype* (Dawkins, 1982), the unit of selection extends beyond the organism: the phenotype of a gene includes all its effects on the world, including artifacts, behaviors, and social structures. The same logic applies to memes, units of cultural replication. A successful meme is not necessarily the most "true" or the most "good": it is the one that replicates with the highest fidelity, fecundity, and longevity in the given environment.

Fictive kinship vocatives are low-entropy memes: short, effortlessly replicable, nearly impossible to misinterpret. Their transmission fidelity is extremely high. Their fecundity is maximal, given that they are inserted into virtually every social exchange. Their longevity is notable: "hermano" as a vocative in Castilian has at least five centuries of documented use in religious and guild contexts. The vocative is, in Dawkinsian terms, a *successful replicator* whose persistence demands functional explanation, not merely historical description.

Dennett's intentionality hierarchy is operationally relevant here. The cultural system that produces and reproduces these vocatives exhibits *first-order intentionality*: it operates as if it had an objective (reducing social friction), but that objective is not conscious for the individual speaker. The Venezuelan who says "primo" does not reason: "I will now activate my interlocutor's kin selection module." The meme uses him.

2.2. Cultural Psychology and the WEIRD/Non-WEIRD Distinction

Henrich, Heine, and Norenzayan (2010) and Henrich's subsequent work in *The WEIRDest People in the World* (2020) demonstrate that human psychology is not universally homogeneous. WEIRD populations (Western, Educated, Industrialized, Rich, Democratic) exhibit cognitive and relational traits that are statistically atypical in the global sample, among them: higher trust in abstract, impersonal institutions; greater tolerance for transactions with strangers; stronger individual vs. group orientation.

Henrich's account of WEIRD psychology's origins is directly relevant. The Western Church's Marriage and Family Program (4th-11th centuries) systematically dismantled intensive kinship networks (clans, lineages, exogamous tribes) across Western Europe. By destroying those networks, it created an institutional vacuum filled by impersonal structures: guilds, corporations, markets, states. Trust migrated from kinship to abstract norm. The Church itself extended fictive kinship through the compadrazgo institution but simultaneously prohibited marriage between compadres, godparents, and godchildren — an explicit recognition that fictive kinship activates the same psychological modules as genetic kinship (Henrich, 2020).

In societies where that transition did not occur, or where abstract institutions subsequently collapsed, trust remains or returns to the relational register. Henrich documents this quantitatively: populations with stronger impersonal prosociality exhibit higher per-capita GDP, more effective governments, and lower corruption. The inverse holds: where institutional quality is low, relational trust substitutes, and fictive kinship devices proliferate. When state institutions collapse, societies rapidly decompose into their most psychologically stable components — tribes, clans, residential communities — which reassume the protective functions the state had appropriated (Henrich, 2020).

2.3. Kin Selection and the Pre-Rational Module

Hamilton's (1964) formalization of kin selection established that organisms evolved specific mechanisms for detecting and preferentially cooperating with genetic relatives. The key parameter is the coefficient of relatedness r: altruistic behavior is evolutionarily stable when $rb > c$, where b is the benefit to the recipient and c is the cost to the actor. Those mechanisms were calibrated in the environment of evolutionary adaptedness (EEA), where statistical co-occurrence of kinship address terms and actual kinship was very high. The system was not designed to detect fictive kinship because fictive kinship was presumably rare or absent in the relevant evolutionary context.

The fictive kinship vocative exploits this calibration mismatch. It activates the kin-detection module before the rational filter can evaluate the signal. This is why the vocative produces behavioral effects despite being, in the standard game-theoretic sense, cheap talk: it operates at the pre-rational level where cheap talk theory does not apply. Section 5 develops this argument in full.

2.4. Multilevel Evolutionary Game Theory

EGT studies how strategies are selected as a function of their relative fitness in repeated interactions. In the multilevel framework of Wilson and Sober (1994), selection operates simultaneously on individuals and groups: a strategy may be individually costly but collectively beneficial, and can fix if between-group competition is sufficiently intense.

Fictive kinship vocatives can be analyzed as strategies in a signaling game. The central prediction is: the frequency of fictive kinship vocatives in speech directed at strangers should covary negatively with institutional quality. This prediction is, in principle, falsifiable with comparative sociolinguistic data. Section 6 develops the partial formalization.

3. Ethnographic Evidence: The Mechanism Documented

Before analyzing the contemporary cases, it is worth establishing that the mechanism proposed is not a theoretical construction: it is directly documented in ethnographic fieldwork across multiple populations.

3.1. The Ju/'hoansi: Vocative Networks as Kin Extension Technology

Among the ju/'hoansi of the Kalahari desert, a flexible name-sharing system uses kinship vocatives to incorporate strangers into the relational network. If a man meets a young woman who shares the name of his daughter, he can ask her to call him "father." She is immediately treated as a daughter — which also places her outside his marriage pool. As Henrich (2020) documents, *"such use of address terms brings people psychologically and socially closer and provides a flexible mechanism for interconnecting everyone within a kinship network."* By one estimate, these naming networks extend to radii of one to nearly two hundred kilometers.

The ju/'hoansi are made uncomfortable by the presence of someone who falls outside their kinship circle: without locating a newcomer in their relational network, they do not know how to behave toward that person. The fictive kinship vocative is not optional social decoration — it is the prerequisite for any cooperative interaction with a stranger.

3.2. The Matsiguenka: A Society Without Personal Names

The traditional Matsiguenka of Amazonia historically lacked personal names entirely. Everyone was addressed exclusively through kinship vocatives: "brother," "mother," "uncle." Identity was fully encoded in the relational register. It was not until the 1950s that American missionaries assigned Spanish names drawn from a Lima telephone directory to Matsiguenka settled in permanent villages (Henrich, 2020). The implication is significant: the kinship vocative is not an embellishment layered on top of an identity system built on personal names. In the Matsiguenka case, *it was the identity system*.

3.3. The Ache: Ritual Kinship Outperforms Genetic Kinship

Among the Ache hunter-gatherers of Paraguay, ritual relationships established during birth and puberty rites create lifelong bonds between adults and children's families. Each ritual relationship is associated with specific roles and generates rights and obligations of mutual support governed by social norms. Hill et al. (2014), cited in Henrich (2020), demonstrate quantitatively that *the Ache have twice as many ritual companions as close genetic relatives outside their own band*, and that ritual relationships are more strongly predictive of food sharing, information sharing, and assistance during illness or injury than close genetic kinship is. The ritual mechanism is not a weak substitute for genetic kinship: it is, in terms of cooperative outcomes, more powerful.

This finding has a direct implication for the paper's argument: the activation of kin-detection modules through ritual and linguistic means is not a second-best approximation of genetic kinship. In cooperative terms, it can outperform it. The meme is not parasitizing the module — it is deploying it more effectively than its original function required.

4. The Formalization Gradient: From Colloquial Vocative to Compadrazgo

One of the most analytically productive features of the fictive kinship vocative phenomenon is that it does not present itself in a single form. It exists along a gradient of formalization, from the momentary and costless to the ritualized and lifelong. Understanding this gradient is essential for the EGT analysis.

4.1. The Gradient Structure

At the minimal end of the gradient: "primo" or "hermano" addressed to a stranger. The cost to the sender is essentially zero. There is no ritual, no third party, no codified obligation. The activation of the kin-detection module is momentary and interaction-specific. If the receiver does not reciprocate, no social norm is violated.

At the intermediate level: "brother" in the African American church and community context. The cost is higher: sustained membership in a community that enforces the norm, behavioral consistency with the implied solidarity, social sanction for defection. The obligations are not formally codified but are enforced by dense reputational networks.

At the maximal end: the Catholic compadrazgo institution. The entry cost is explicit and ritual: a ceremony, a public commitment, the presence of the child as anchor of the relationship. The obligations are lifelong, socially exigible, and extend to the godchild as well as to the relationship between the two adults (compadres). The Church simultaneously recognized and constrained the institution: it extended fictive kinship through compadrazgo but prohibited marriage between compadres, godparents, and godchildren, acknowledging that the psychological modules activated are identical to those activated by genetic kinship.

The colloquial "compa," used widely in Central America and Mexico, is the linguistic trace of the compadrazgo institution operating in informal mode. It invokes the normative framework of compadrazgo — its obligations, its solidarity, its claim on reciprocity — without paying the entry cost. In Betzig's (1986) terms, it is an attempt to access a *reservoir of potential reciprocity* by linguistic means alone. Whether that attempt succeeds depends on whether the receiver accepts the implied frame.

4.2. Theoretical Implications of the Gradient

The gradient structure has three implications for the paper's argument.

First, it confirms that the mechanism is real and not metaphorical. The compadrazgo institution demonstrates that human societies have, in multiple independent cultural lineages, constructed formal institutions around exactly the cognitive mechanism the colloquial vocative exploits informally. The formal institution validates the hypothesis about the informal one.

Second, the gradient allows a partial test of the cheap talk problem. At the minimal end (colloquial "primo"), the signal is pure cheap talk. At the maximal end (compadrazgo), the signal carries a genuine cost. The fact that cooperative effects are observed at both ends, though stronger and more durable at the costly end, is consistent with the pre-rational activation hypothesis: even the costless version works because it triggers a module, not because it is a credible rational signal.

Third, the gradient maps directly onto the three functions identified in Section 5. Horizontal leveling is the dominant function at the minimal end. Clientelar lubrication is most visible at the intermediate level, where the vocative implies obligation without formal enforcement. The full compadrazgo institution is where clientelar capture is most historically documented, particularly in Latin American political machines where godparent networks were the structural basis of electoral clientelism.

5. Three Functions in Tension

5.1. Horizontal Leveling

Boehm (1999) documents that hunter-gatherer groups exhibit active leveling mechanisms: subordinates collectively sanction individuals who attempt to accumulate disproportionate power or resources. Leveling is an evolutionarily stable strategy in small groups with high interdependence. Fictive kinship vocatives are linguistic leveling instruments: by addressing a stranger as "cousin," the speaker signals non-predation and acknowledges equal status for the purposes of the interaction. It is preemptive verbal *leveling*.

This function is directly documented in the ju/'hoansi case: incorporating a stranger into the kinship network through a vocative is literally a mechanism for ensuring that the group includes everyone present within a circle of relatedness, because without that inclusion they do not know how to behave cooperatively toward the outsider. The leveling function is not a secondary interpretation — it is the primary adaptive rationale.

5.2. Clientelar Lubrication

The second function is, paradoxically, the opposite: the same vocative can serve as the preamble to a request that creates asymmetric reciprocity debt. "Compa, do me this favor" does not signal equality — it invokes a normative frame of mutual obligation in order to extract a unilateral benefit. The meme lubricates the very clientelar order it superficially appears to resist.

Betzig (1986) defines wealth as a *"reservoir of potential reciprocity"* and friendship as *"just another word for it."* The fictive kinship vocative is an attempt to access that reservoir by linguistic means: to claim the social credit of an established relationship without having paid its construction costs. In dense clientelar networks, this operates as what might be called a *cooperation trap*: the vocative induces the receiver to lower their guard by activating the kin selection module, creating a vulnerability that the sender can exploit.

The compadrazgo institution makes this function most visible historically. In Latin American political machines from the colonial period onward, godparent networks were the structural backbone of electoral clientelism: the compadre relationship created obligations that could be called in at voting time, in exchange for protection, employment, or access to state resources. The kinship language encoded extraction in the vocabulary of solidarity.

5.3. Semantic Hollowing Through Appropriation

The third function emerges from the "bro" case. When a meme originating in institutional exclusion is appropriated by a group with high institutional access, it can shed its original adaptive payload and mutate toward cultural identity signaling. The *"bro" of the North American university fraternity* is functionally distinct from the *"brother" of the Black church in Chicago*. They share the replicator but not the phenotype.

This process of semantic hollowing is predictable from the extended phenotype framework: when the environment changes (from institutional exclusion to inclusion), the meme survives through cultural inertia but its original adaptive function disappears. What remains is a semantic shell available for colonization by any function the new environment makes salient. In the "bro" case, that function became masculinity performance and peer-group signaling — what became known as "bro culture." This is an instance of *memetic parasitism*: the replicator persists in the cultural pool not because it serves its original function but because it has anchored to a different one that guarantees continued replication.

The three functions coexist and compete within the same replicator. The same utterance can perform all three simultaneously depending on the interactional context, the institutional ecology, and the degree of formalization. This ambivalence is a property of the meme, not an artifact of the analysis.

6. The Cheap Talk Problem and Its Resolution

The hypothesis generates a theoretical problem that any reviewer in behavioral biology or EGT will raise immediately. Under Farrell and Rabin's (1996) cheap talk framework, a signal with near-zero cost should not be credible at equilibrium: rational receivers should discount it entirely. If the fictive kinship vocative is *cheap talk*, why does it have behavioral effects?

The resolution lies in the distinction between rational-strategic and pre-rational processing. The vocative does not operate at the level where cheap talk theory applies. It activates a cognitive module that processes the signal *before* the rational filter can discount it. Hamilton's (1964) kin selection mechanisms were calibrated in the EEA, where statistical co-occurrence of kinship terms and actual kinship was very high. The system was not designed to detect fictive kinship because fictive kinship was presumably rare or absent in that context.

The ju/'hoansi evidence is directly relevant here: incorporating a stranger via a kinship vocative immediately changes how that person is treated, including placing them outside the marriage pool. This is not a rational calculation — it is an automatic module response. The vocative exploits a calibration mismatch between the ancestral environment and the current one.

The gradient structure provides additional evidence. At the minimal end (colloquial "primo"), the signal is cheap but still produces cooperative effects. At the maximal end (compadrazgo), the same mechanism is formalized precisely because cheap activation alone is insufficient for durable cooperation: the institution adds cost to increase credibility. The fact that both ends of the gradient produce cooperative effects — though of different durability — is consistent with pre-rational activation at the base and rational reinforcement through cost at the formal end.

This mechanism has implications beyond sociolinguistics. Political speeches, religious rhetoric, and clientelar discourse that systematically deploy kinship vocabulary are exploiting the same cognitive module at scale. The analysis of vocatives is a tractable entry point into a larger theory of institutional manipulation through kinship language.

7. Partial Formalization in EGT

The central hypothesis can be stated as a falsifiable EGT prediction:

> *Let I be an index of institutional quality (1 = maximum impersonal trust, 0 = institutional collapse). Let V be the frequency of fictive kinship vocatives directed at strangers in a given population. The hypothesis predicts a negative correlation between I and V: as institutional quality decreases, the fitness of the vocative as a coalitional signaling strategy increases.*

Henrich (2020) provides partial parametrization: populations with higher impersonal prosociality exhibit higher per-capita GDP, more effective governments, and lower corruption. The inverse — lower institutional quality correlating with higher reliance on relational trust mechanisms — is consistent with the prediction. Full parametrization requires cross-national corpus data on vocative frequencies correlated with institutional quality indices (Rule of Law Index, World Values Survey interpersonal trust measures). That data is not yet systematically available, which constitutes a primary limitation of this paper and a *research agenda item* rather than a disqualifying gap.

Table 2. Signaling game payoff matrix (ordinal)

	Receiver uses vocative	**Receiver does not use vocative**
Sender uses vocative	Low-cost cooperation (3, 3)	Exploitation risk for sender (1, 4)
Sender does not use vocative	Receiver advantage (4, 1)	Neutral stranger interaction (2, 2)

In a low-I environment, the payoff from mutual low-cost cooperation (3,3) is systematically superior to neutral stranger interaction (2,2), and the exploitation risk (1,4) is more tolerable than unilateral disadvantage (4,1). This generates selection pressure toward mutual vocative use as a Nash equilibrium. In a high-I WEIRD environment, the neutral interaction payoff (2,2) is sufficient because abstract institutions guarantee the contract, and in-group signaling toward strangers may be actively penalized. This accounts for the differential *fitness* of the meme across institutional environments.

The gradient structure adds a second dimension to the formalization: as cooperation requirements extend beyond a single interaction, agents move along the gradient from colloquial activation toward more costly and durable signals. The compadrazgo institution represents the equilibrium point where the cost of the signal is sufficient to sustain long-term cooperative relationships that the informal vocative alone cannot maintain.

8. Implications and Research Agenda

8.1. The Vocative as Institutional Thermometer

If the hypothesis is correct, the frequency of fictive kinship vocatives directed at strangers is a linguistic indicator of institutional temperature. Not the only indicator, nor the most precise, but one of the most economical to collect: corpus-based sociolinguistic analysis, without surveys or experimental design.

This prediction connects cognitive linguistics with institutional economics. The informal social capital that Putnam (1993) and others measure through trust surveys may have a directly observable linguistic correlate in vocative patterns. A cross-national corpus study correlating vocative frequencies with institutional quality indices would provide the empirical test the hypothesis requires. The PRESEEA corpus (Proyecto para el Estudio Sociolingüístico del Español de España y América), which documents spontaneous speech across Latin American cities with varying institutional quality, is a candidate data source.

8.2. Compadrazgo as a Natural Experiment

The compadrazgo institution provides an unusual natural experiment for the hypothesis. Because it is a formally codified fictive kinship institution with documented historical variation in density and function across Latin American societies, it allows a diachronic test: do populations with historically denser compadrazgo networks exhibit lower institutional development in subsequent periods, controlling for other variables? Or does the direction of causation run the other way — do populations with weaker institutions develop denser compadrazgo networks as a substitute?

The causal direction question is not resolvable from the theoretical argument alone. But it is empirically tractable with historical data on compadrazgo density (available from parish records across Latin America) correlated with institutional quality measures. That analysis has not been performed.

8.3. Implications for Institutional Design

The cheap talk resolution proposed in Section 6 has implications beyond sociolinguistics. If fictive kinship vocatives work by exploiting a pre-rational kin detection module, then the same mechanism is available to any agent deploying kinship language at scale: populist political discourse framing the polity as a family or brotherhood; religious institutions using "brother" and "sister" as universal address terms; military organizations deploying "band of brothers" rhetoric; clientelar political machines encoding extraction in the language of reciprocal obligation.

In each case, the cognitive mechanism is identical: a low-cost signal activates a pre-rational module calibrated for a different ancestral environment, producing cooperative behavior that benefits the sender disproportionately. Understanding the vocative case at the micro-interactional level provides a tractable model for the broader phenomenon of institutional manipulation through solidarity language.

8.4. Limitations

Three limitations deserve explicit acknowledgment. First, the comparative empirical evidence is insufficient: the hypothesis is plausible and falsifiable, but has not been tested with systematic vocative frequency data. The ethnographic cases establish the mechanism; they do not establish the predicted cross-national frequency distribution. Second, the EGT formalization is illustrative: ordinal payoffs cannot substitute for parametrized models, and the gradient structure requires a more complex game-theoretic treatment than the binary matrix presented here. Third, the analysis is restricted to three cases drawn from a limited geographic and linguistic range; the hypothesis requires testing in typologically distinct languages and kinship systems.

These limitations place the paper correctly: as a research program in initial state, with falsifiable predictions and a clear empirical agenda.

9. Conclusion

"Primo," "compadre," and "brother" are not merely words. They are social regulation technologies of varying cost, selected for their capacity to activate evolutionarily conserved cognitive modules in environments where abstract institutions fail to produce sufficient impersonal trust. Their geographic distribution, formalization gradient, and diffusion trajectories reveal the institutional ecology of the societies that generate them.

The Venezuelan who addresses a stranger as "primo" and the participant in a Catholic compadrazgo ceremony are deploying the same underlying mechanism at radically different points on the formalization gradient. The Black American who says "brother" in a Chicago church is doing the same thing in a context of systemic institutional exclusion. The elite university student who says "bro" to his roommate is reproducing a replicator that shed its original adaptive function and acquired another.

The differences among these cases are not primarily linguistic. They are institutional. And that is precisely what the hypothesis predicts.

The broader methodological implication: vocative patterns are a window into the institutional psychology of a society. Where fictive kinship proliferates in speech directed at strangers, institutions are failing to produce impersonal trust. Where it disappears or mutates, the explanation requires distinguishing between institutional development and semantic hollowing through appropriation. The compadrazgo gradient shows that the same cognitive mechanism can be institutionalized at any level of formalization — from a single word to a lifelong binding commitment — depending on what the cooperative environment requires. Making those distinctions requires exactly the kind of theoretically anchored comparative analysis this paper attempts to initiate.

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